

PULSARPOWER | 2 heaters + 4 monitored DC outlets + 4 USB 2.0 ports

PROTOTYPE FLOORPLAN – routing, full-load thermal validation and new firmware are still required.

Power input

VIN
GND

File: power_input.kicad_sch

DC supplies

VIN
GND
V5
V3

File: supplies.kicad_sch

Heaters

VIN
V5
GND
HTR0
HTR1
V3
SCL
SDA

File: heaters.kicad_sch

Controller

V3
GND
HTR0
HTR1
MCU_DP
MCU_DM
USB_ATTACH
SCL
SDA
DC_EN1
IMON1
DC_PG1
DC_EN2
IMON2
DC_PG2
DC_EN3
IMON3
DC_PG3
DC_EN4
IMON4
DC_PG4

File: controller.kicad_sch

USB hub

V3
GND
MCU_DP
MCU_DM
USB_ATTACH
USB1_DP
USB1_DM
USB1_EN
USB1_OC
USB2_DP
USB2_DM
USB2_EN
USB2_OC
USB3_DP
USB3_DM
USB3_EN
USB3_OC
USB4_DP
USB4_DM
USB4_EN
USB4_OC

File: usb_hub.kicad_sch

DC protection

VIN
V3
GND
DC_EN1
DC_EN2
DC_EN3
DC_EN4
DC_SAFE1
DC_SAFE2
DC_SAFE3
DC_SAFE4

File: dc_protection.kicad_sch

DC outlet 1

VIN
V3
GND
DC_SAFE1
IMON1
DC_PG1

File: dc_outlet.kicad_sch

DC outlet 2

VIN
V3
GND
DC_SAFE2
IMON2
DC_PG2

File: dc_outlet.kicad_sch

DC outlet 3

VIN
V3
GND
DC_SAFE3
IMON3
DC_PG3

File: dc_outlet.kicad_sch

DC outlet 4

VIN
V3
GND
DC_SAFE4
IMON4
DC_PG4

File: dc_outlet.kicad_sch

USB port 1

V5
GND
USB1_DP
USB1_DM
USB1_EN
USB1_OC

File: usb_port.kicad_sch

USB port 2

V5
GND
USB2_DP
USB2_DM
USB2_EN
USB2_OC

File: usb_port.kicad_sch

USB port 3

V5
GND
USB3_DP
USB3_DM
USB3_EN
USB3_OC

File: usb_port.kicad_sch

USB port 4

V5
GND
USB4_DP
USB4_DM
USB4_EN
USB4_OC

File: usb_port.kicad_sch

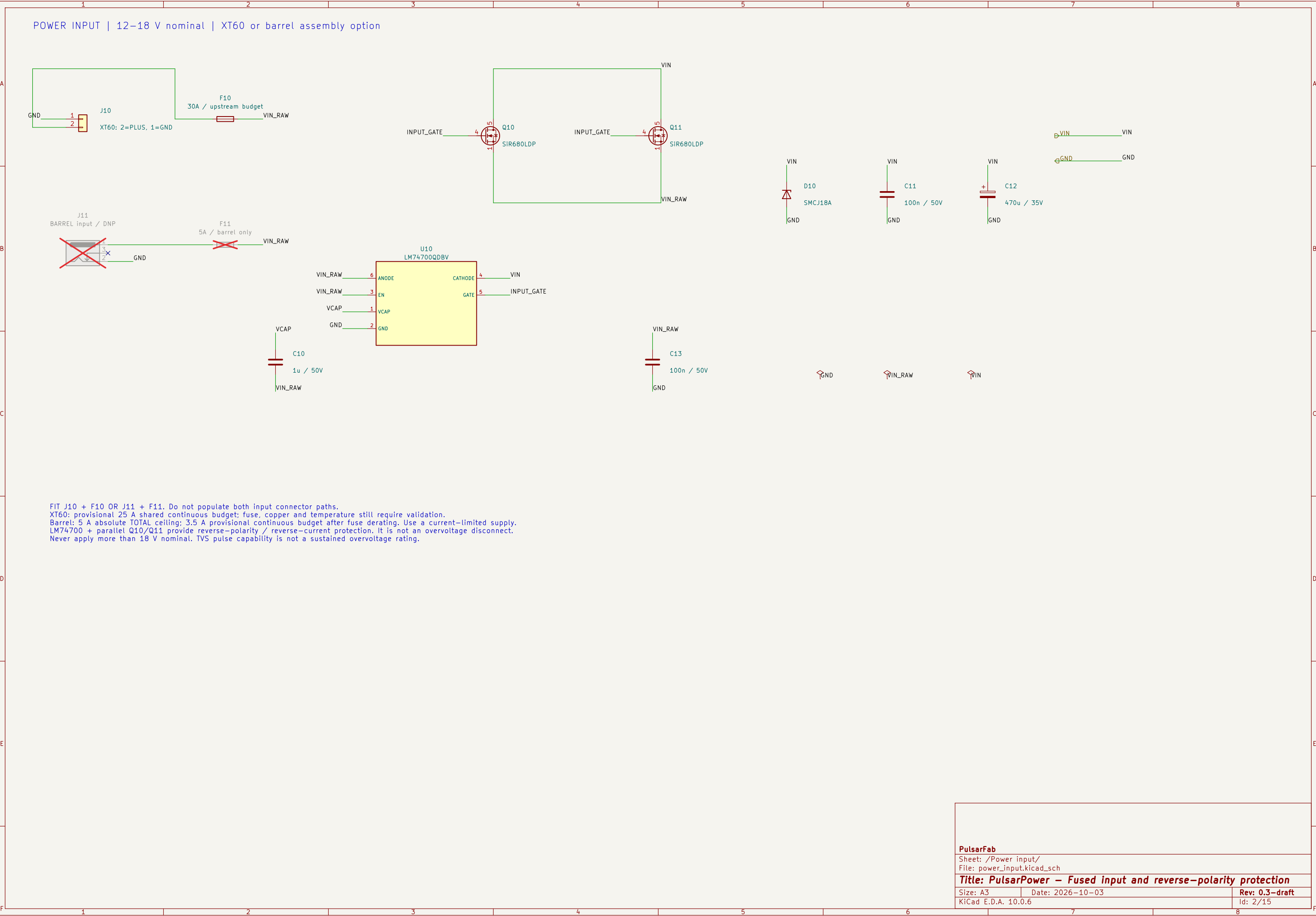
RATINGS TO VALIDATE
12–18 V nominal input
4 x 5 A DC outlet targets
2 x 5 A heater targets
25 A shared XT60 budget
3.5 A continuous barrel budget
USB: 500 mA per port
No PWM on DC outlets

Same ground throughout.
High-current and USB return paths
are separated by placement.
Do not backfeed DC outputs.

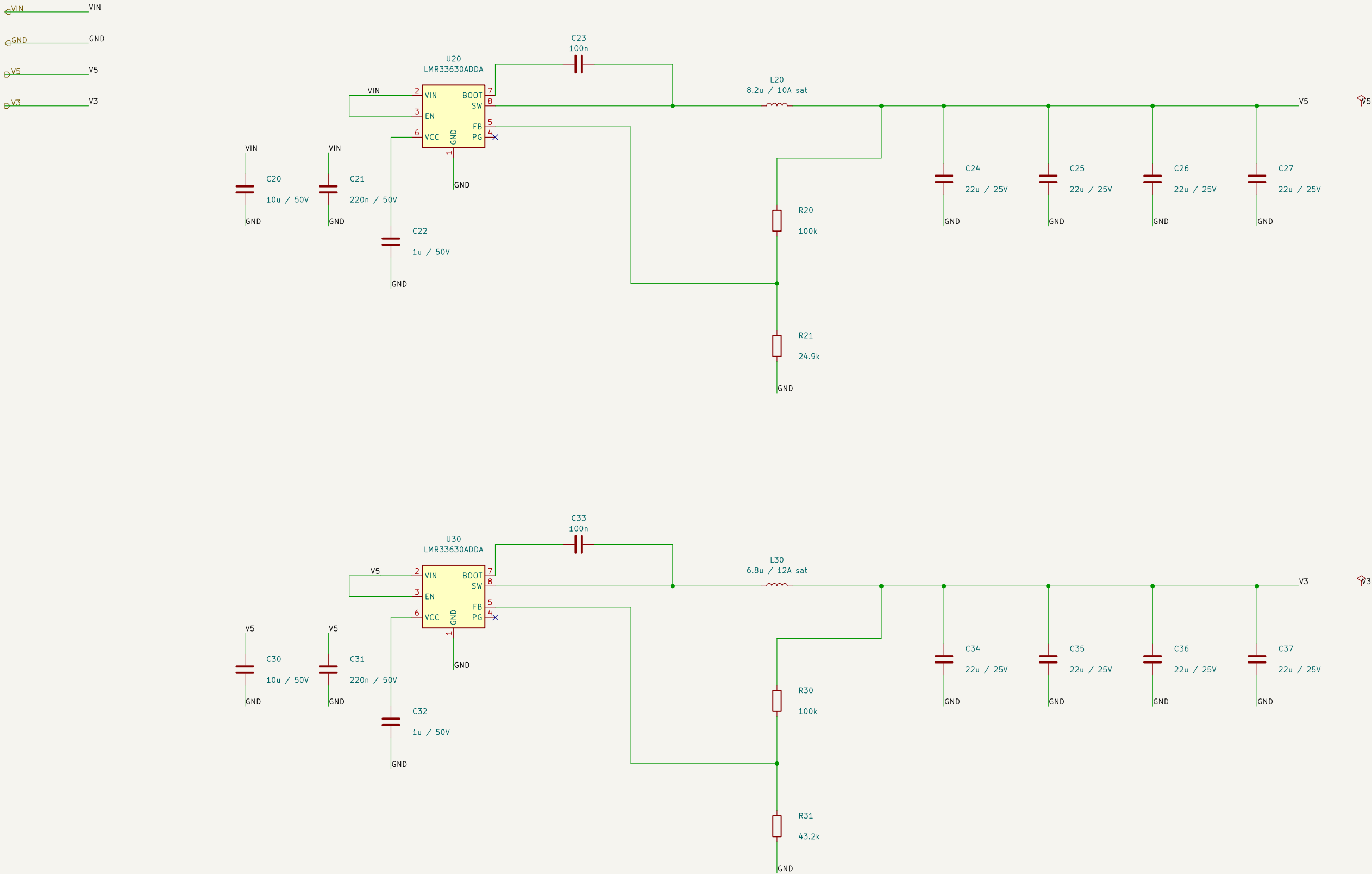
PulsarFab
Sheet: /
File: pulsardewhub.kicad_sch

Title: PulsarPower – System interconnect

Size: A2 Date: 2026–10–03 Rev: 0.3–draft
KiCad E.D.A. 10.0.6 Id: 1/15



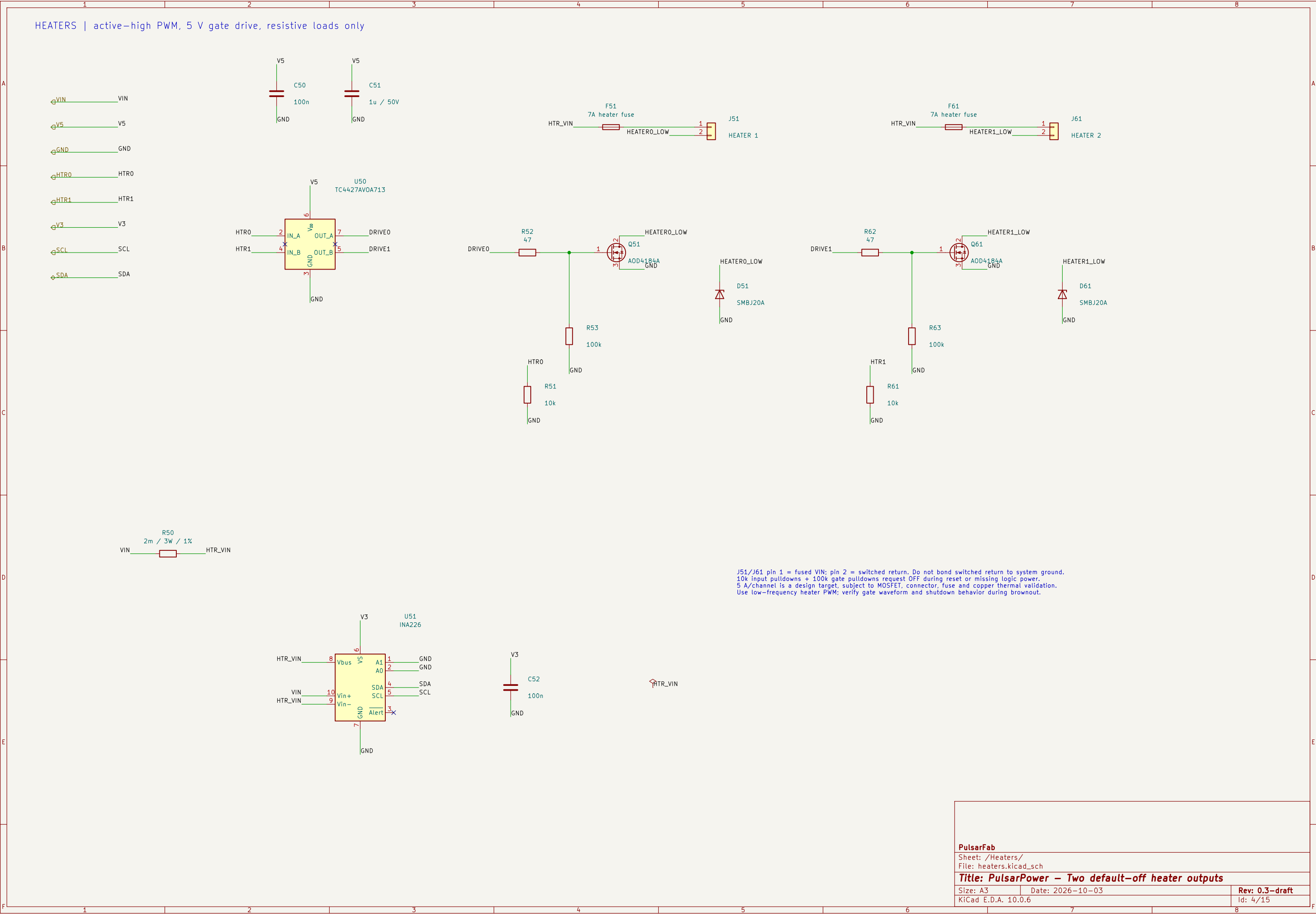
DC SUPPLIES | LMR33630ADDA, 400 kHz | 5 V / 3 A + 3.3 V logic



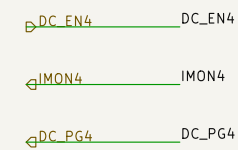
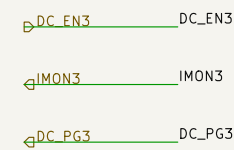
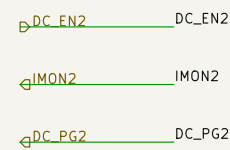
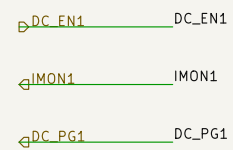
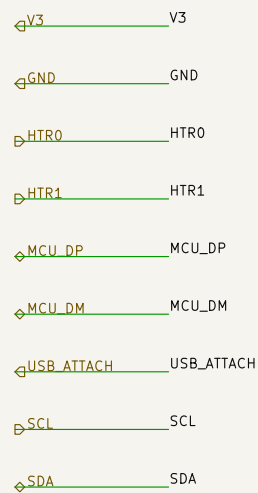
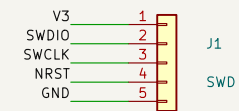
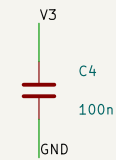
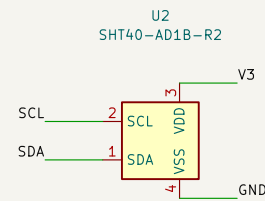
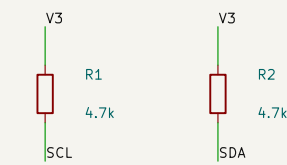
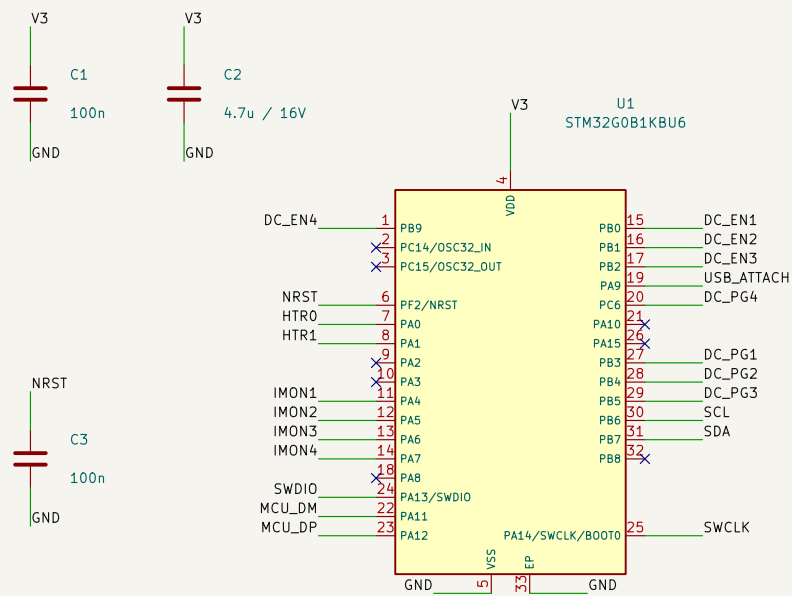
Follow TI SNVSAN3F Table 9–2. 5 V uses nearest preferred 8.2 uH to the 8 uH example.
Place ceramic input/boot/VCC capacitors at the IC; keep SW copper compact and FB away from SW.
USB budget: 4 x 500 mA plus about 0.4 A for logic at full activity. Verify inductor saturation, MLCC bias, transient response and regulator temperature before routing release.

PulsarFab		
Sheet: /DC supplies/ File: supplies.kicad_sch		
Title: PulsarPower – 5 V USB and 3.3 V logic supplies		
Size: A3	Date: 2026–10–03	Rev: 0.3–draft
KiCad E.D.A. 10.0.6		Id: 3/15

HEATERS | active-high PWM, 5 V gate drive, resistive loads only

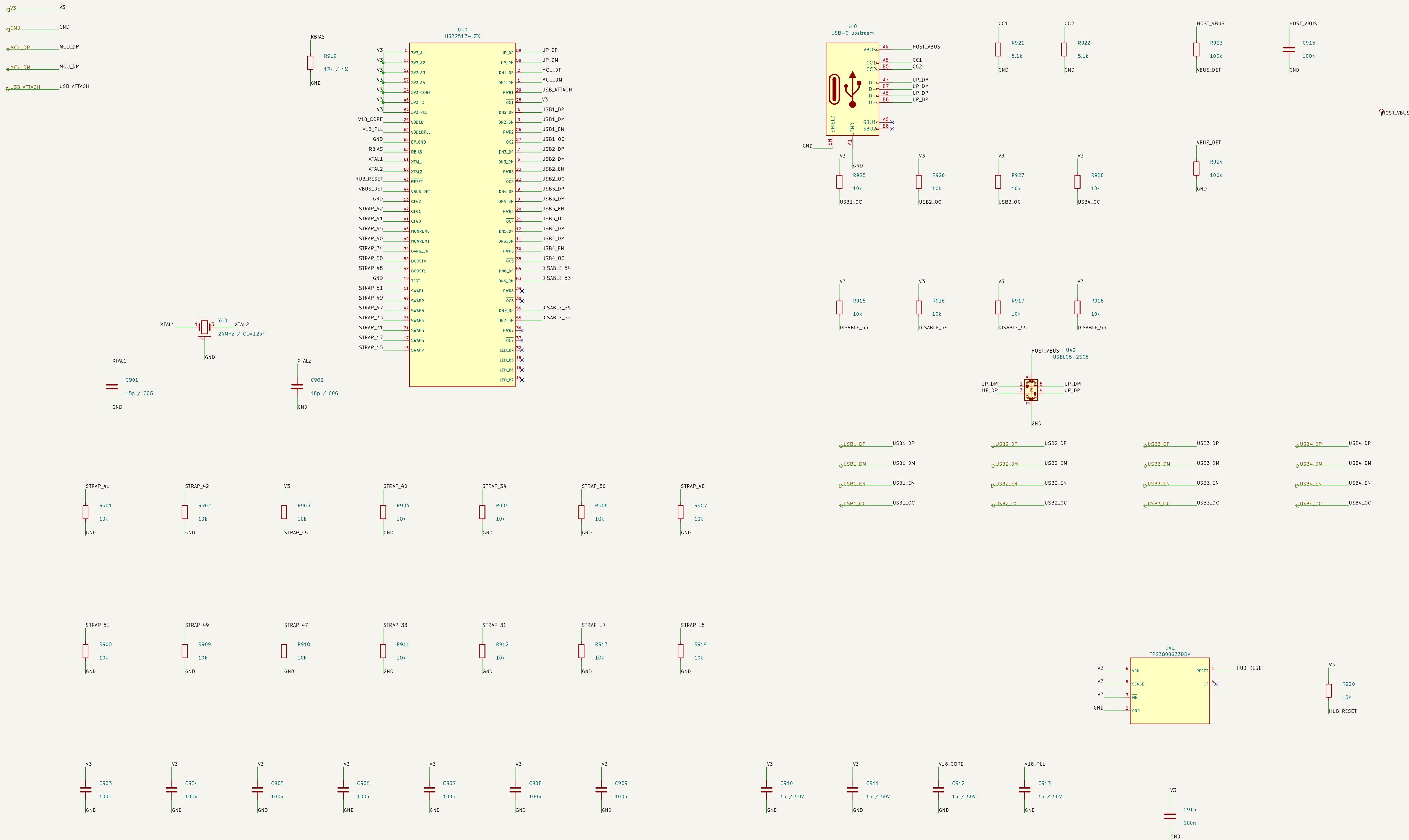


CONTROLLER | 2 heater PWM + 4 DC enables / ADC current readings / faults



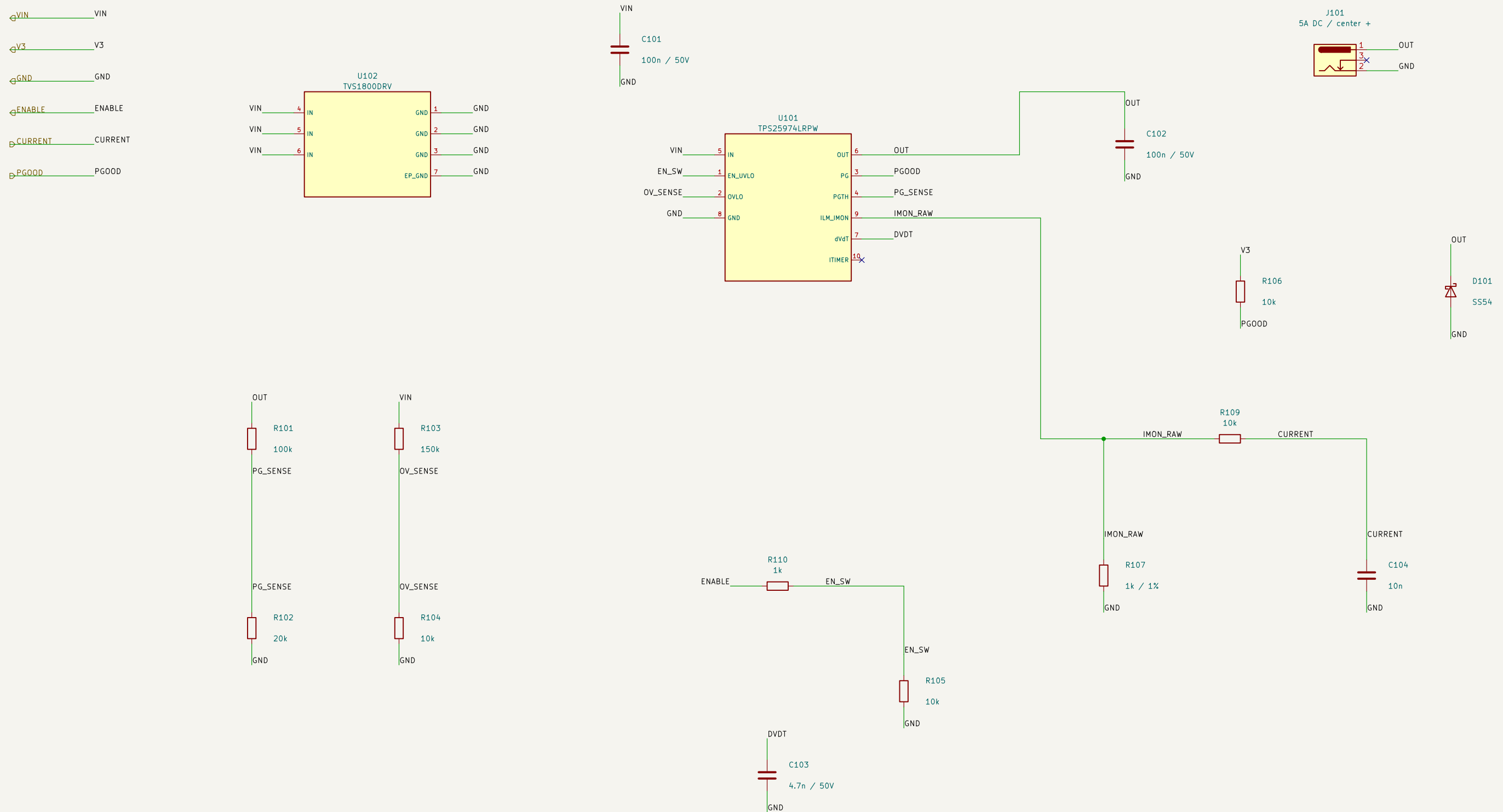
PA0/PA1: active-high TIM2 heater drive. PB0/PB1/PB2/PB9: static DC GPIO only (never PWM).
PA4..PA7: ADC IMON1..4. PB3/PB4/PB5/PC6: active-low DC faults.
PA9: internal hub port 1 enable (3.3 V). Attach USB only while high; disable UCPD dead-battery pull-down.
SHT40 at 0x44: keep physically away from power components. New firmware pin map is required.

USB HUB | USB2517 | internal port 1 + external ports 2-5; ports 6/7 disabled



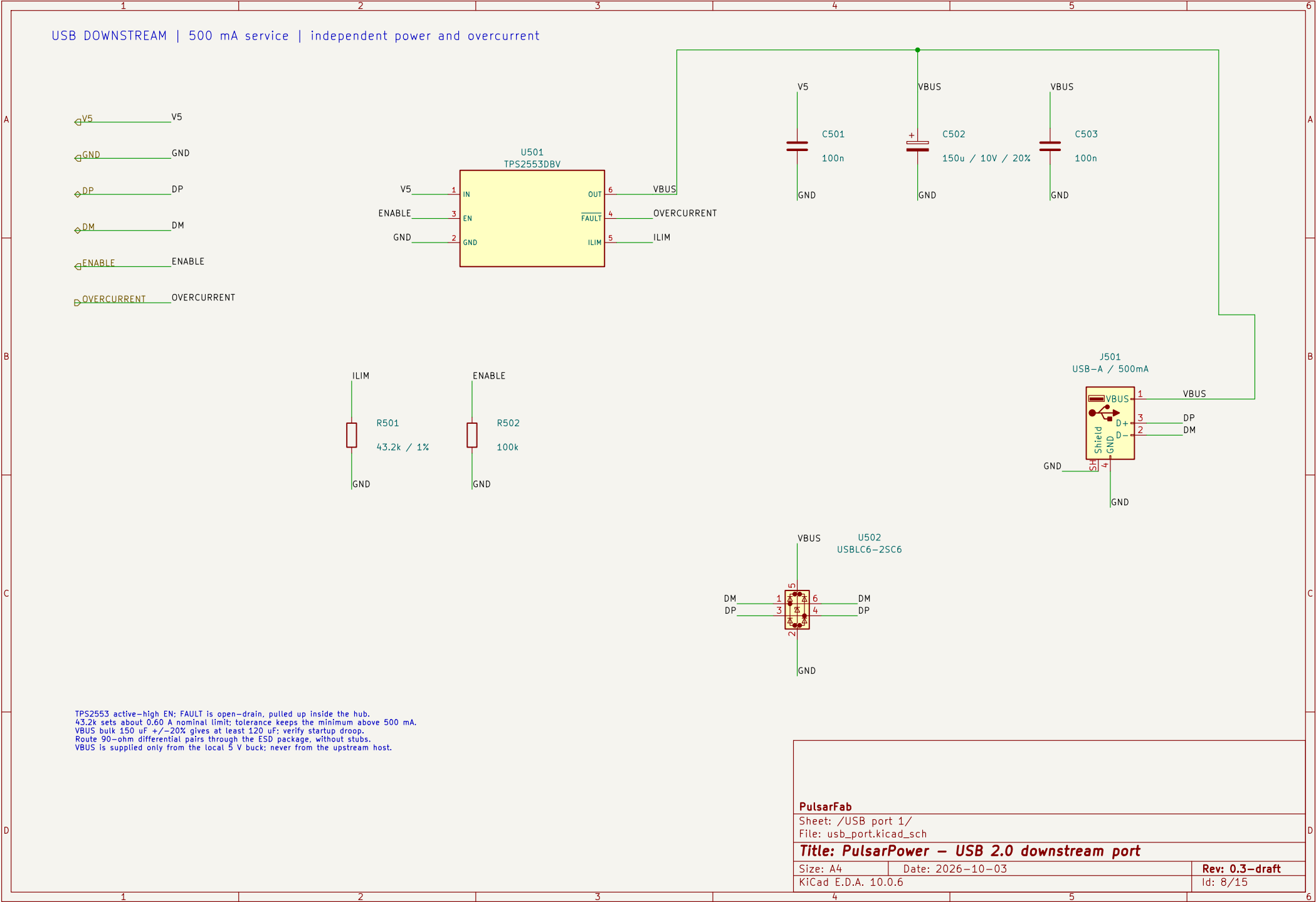
Straps: CFG=000 self-powered; NONREM=01 makes port 1 permanent; individual power/OC; normal polarity. Unused ports 6/7: both D+ and D- pulled to 3V3 through 10k; PWR outputs unused, OC inputs unused. VDD18 and VDD18PLL are separate INTERNAL regulated voltage outputs; capacitors only; do not connect them together. Supervisor: 3.07 V threshold, 20 ms delay. Crystal caps start at 18 pf; verify actual board parasitics/drive level. Keep an uninterrupted ground plane under every USB pair. Verify 90-ohm geometry with the ordered stackup.

SWITCHED DC OUTLET | 12-18 V / 5 A service | hardware voltage inhibit / latched breaker

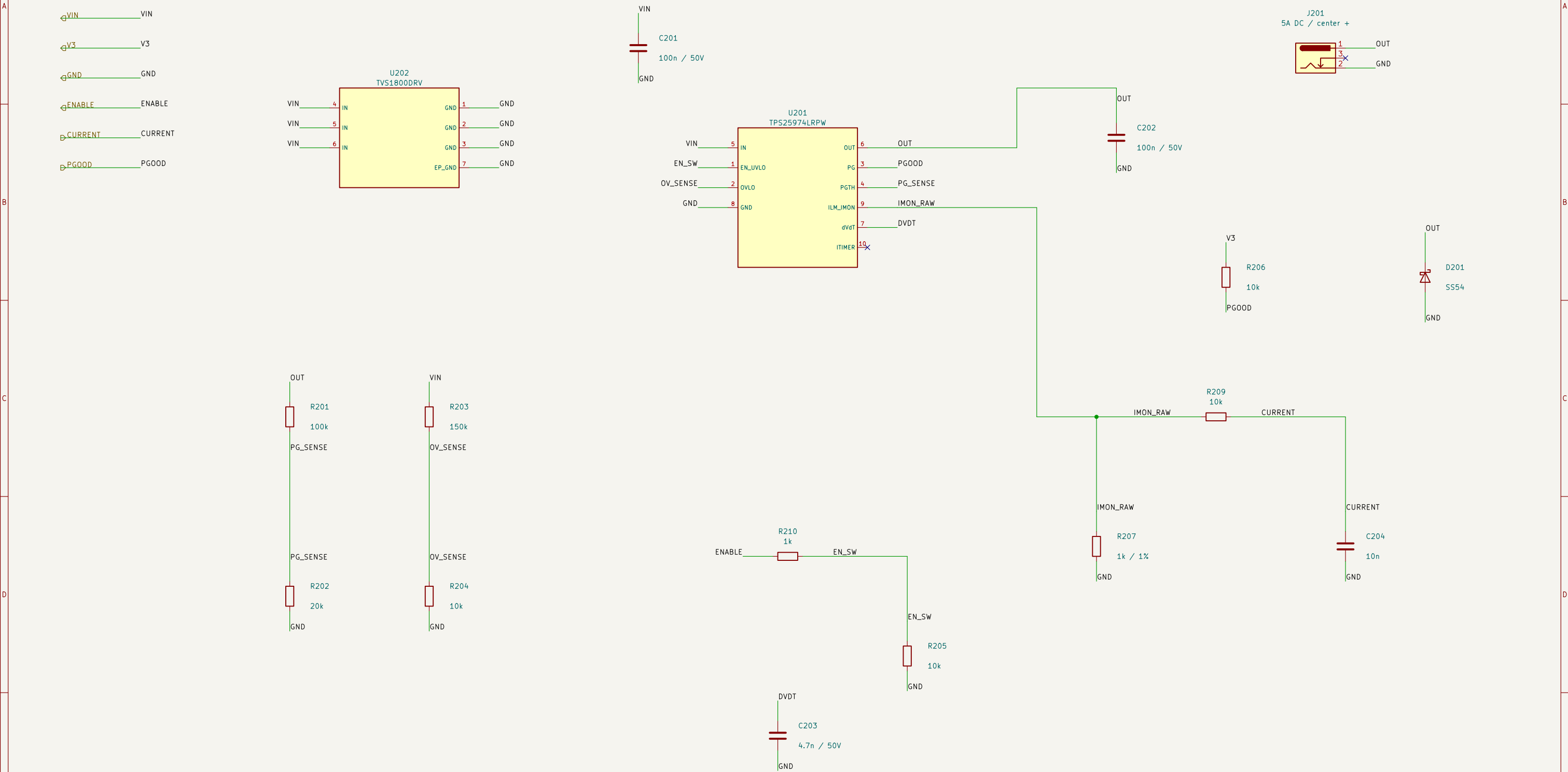


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open; latch-off. ILM is BOTH current setting and monitoring: $105.5 \text{ uA/A} \times 1k = 0.1055 \text{ V/A}$. 0.528 V at 5 A. 100k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM. Independent local OVLO: 150k/10k $\rightarrow$ 19.2 V nominal. PG threshold: 100k/20k $\rightarrow$ 7.2 V nominal. Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs. In pad5 / OUT pad6 are power/thermal; NOT GND. Reserve copper and thermal-via space. TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A $-$ 8 $-$ 20 u / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

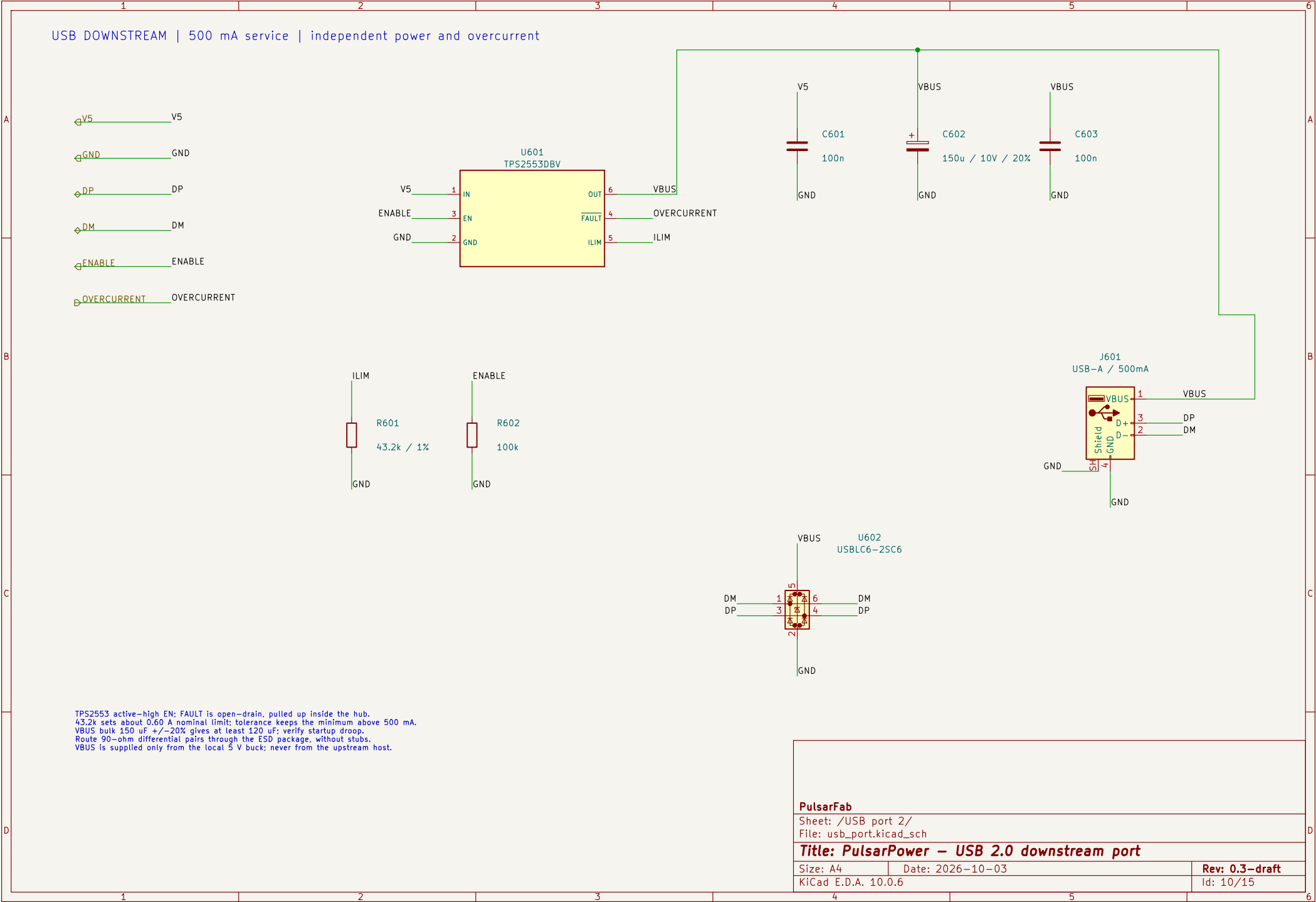


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

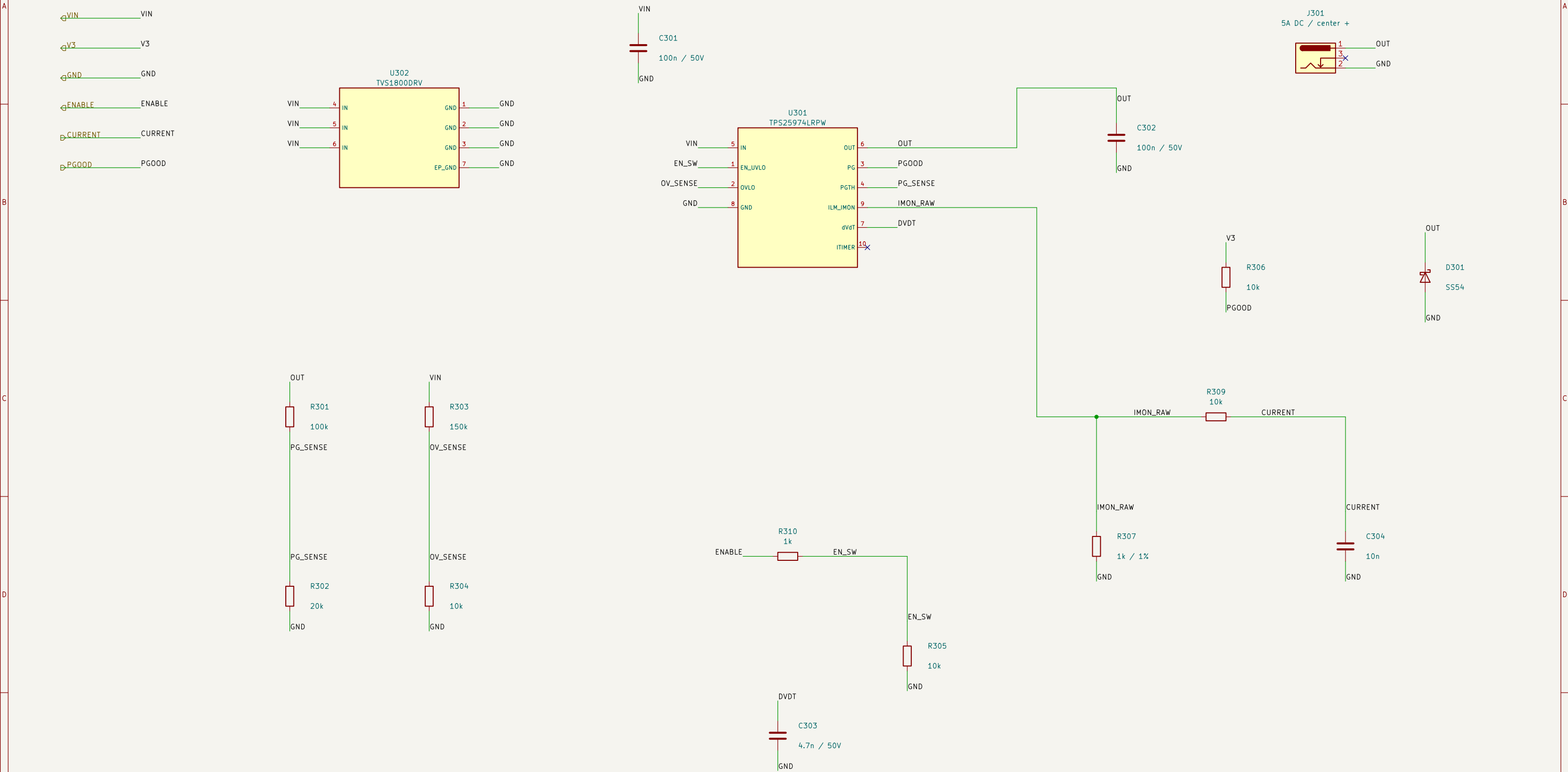


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open: latch-off.
ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM.
Independent local OVLO: 150k/10k -> 19.2 V nominal. PG threshold: 100k/20k -> 7.2 V nominal.
Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs.
IN pad5 / OUT pad6 are power/thermal rails, NOT GND. Reserve copper and thermal-via space.
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USB DOWNSTREAM | 500 mA service | independent power and overcurrent

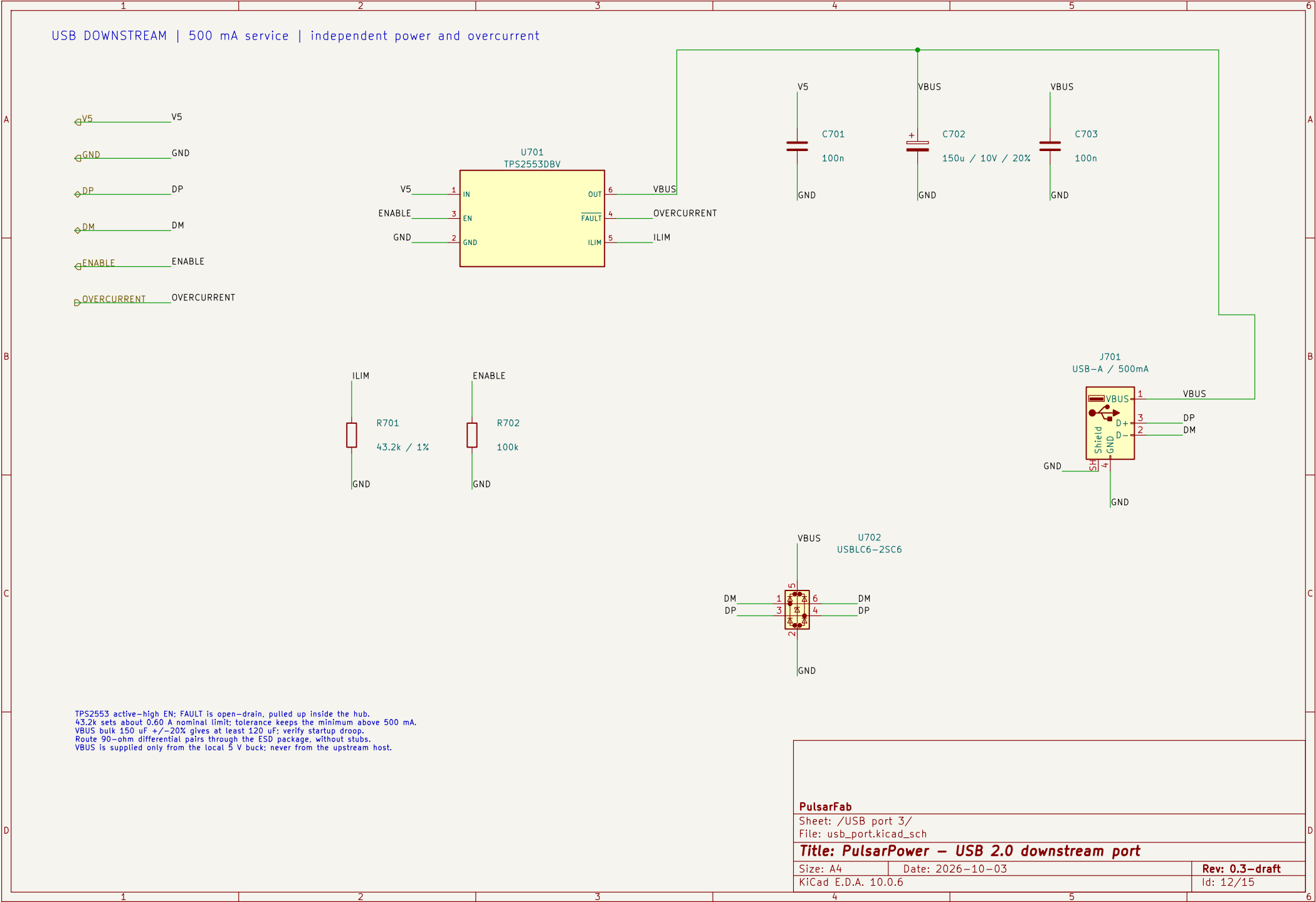


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker



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USB DOWNSTREAM | 500 mA service | independent power and overcurrent



PulsarFab

Sheet: /USB port 3/
File: usb_port.kicad_sch

Title: PulsarPower – USB 2.0 downstream port

Size: A4

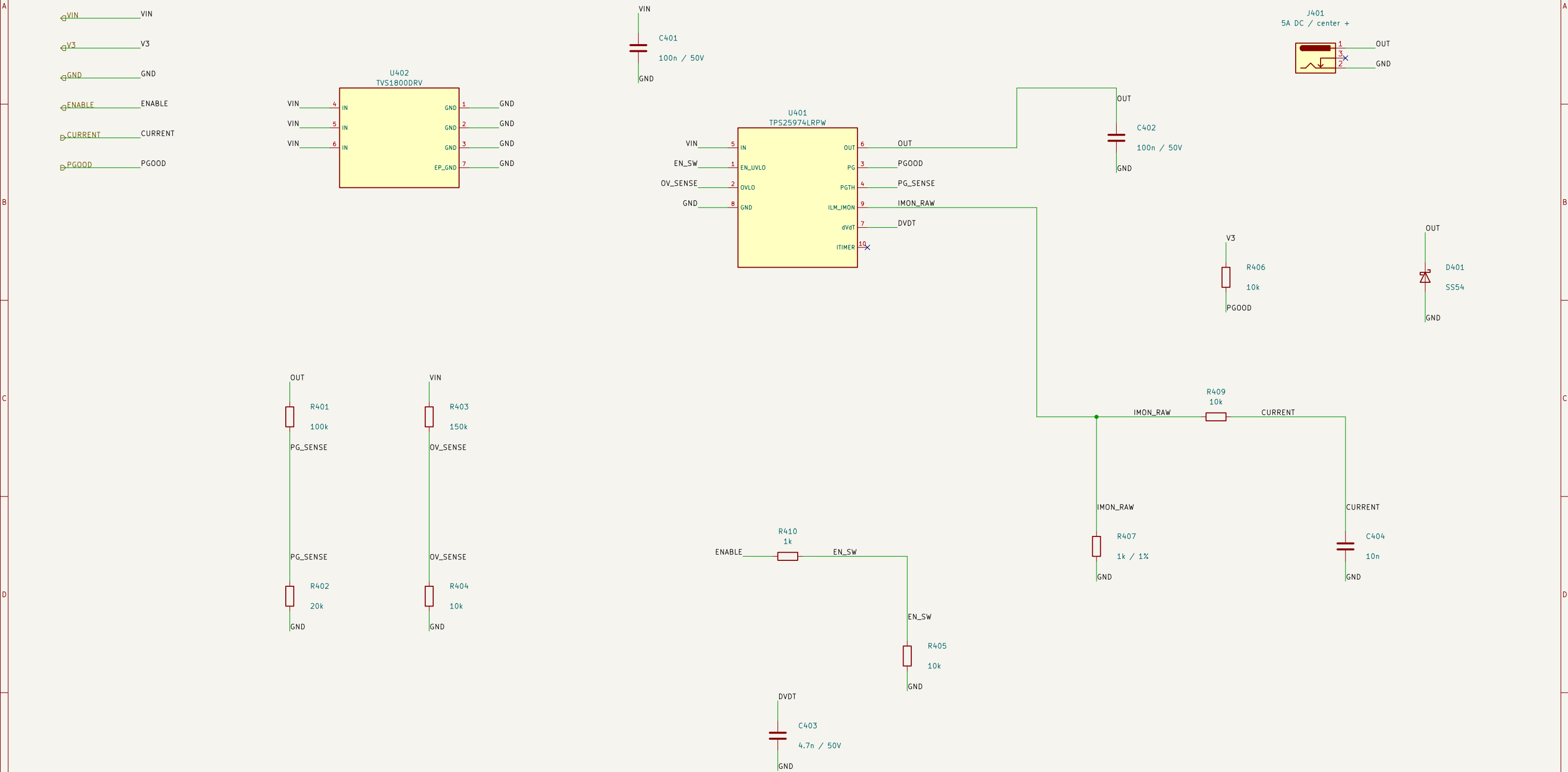
Date: 2026-10-03

Rev: 0.3-draft

KiCad E.D.A. 10.0.6

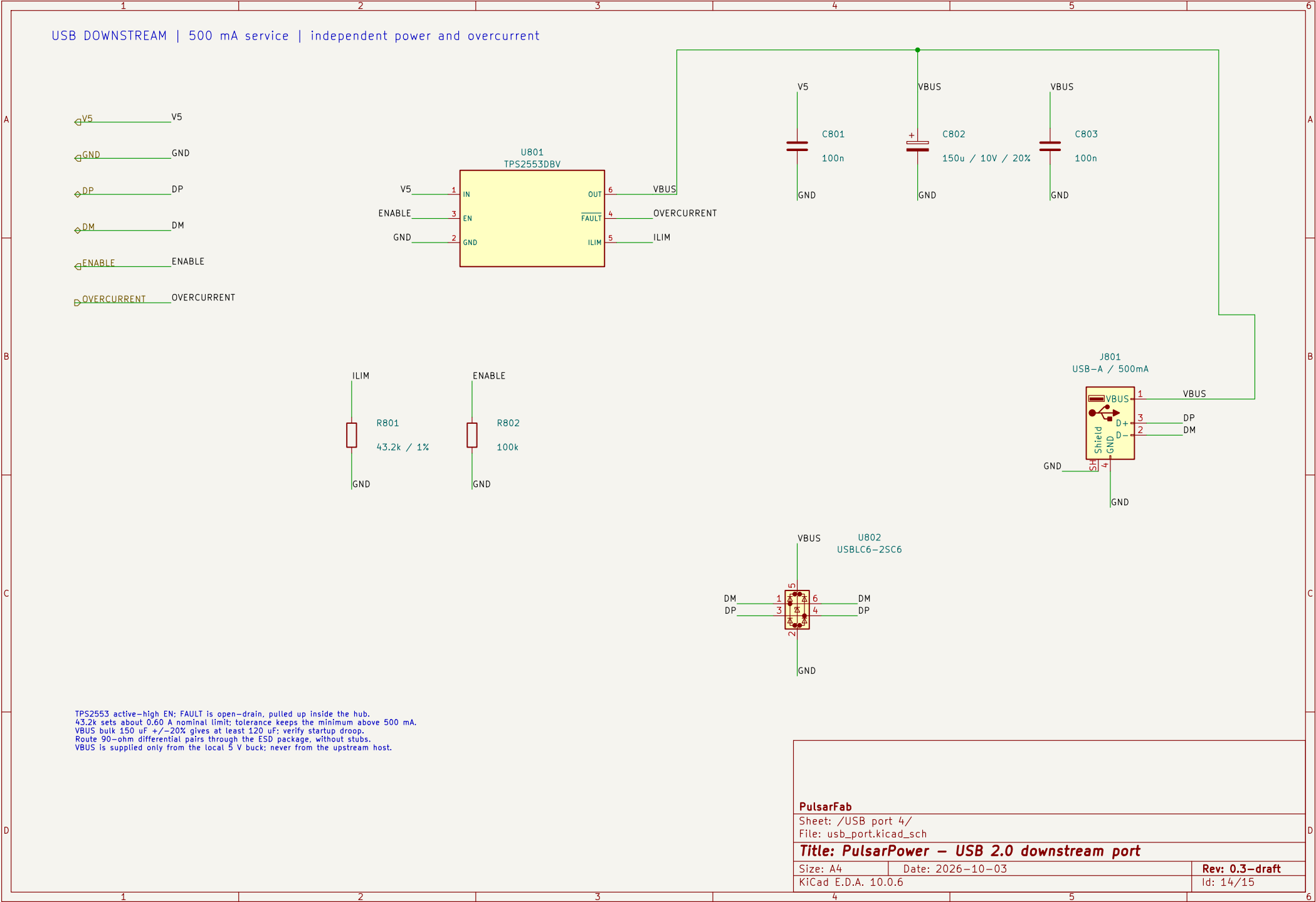
Id: 12/15

SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

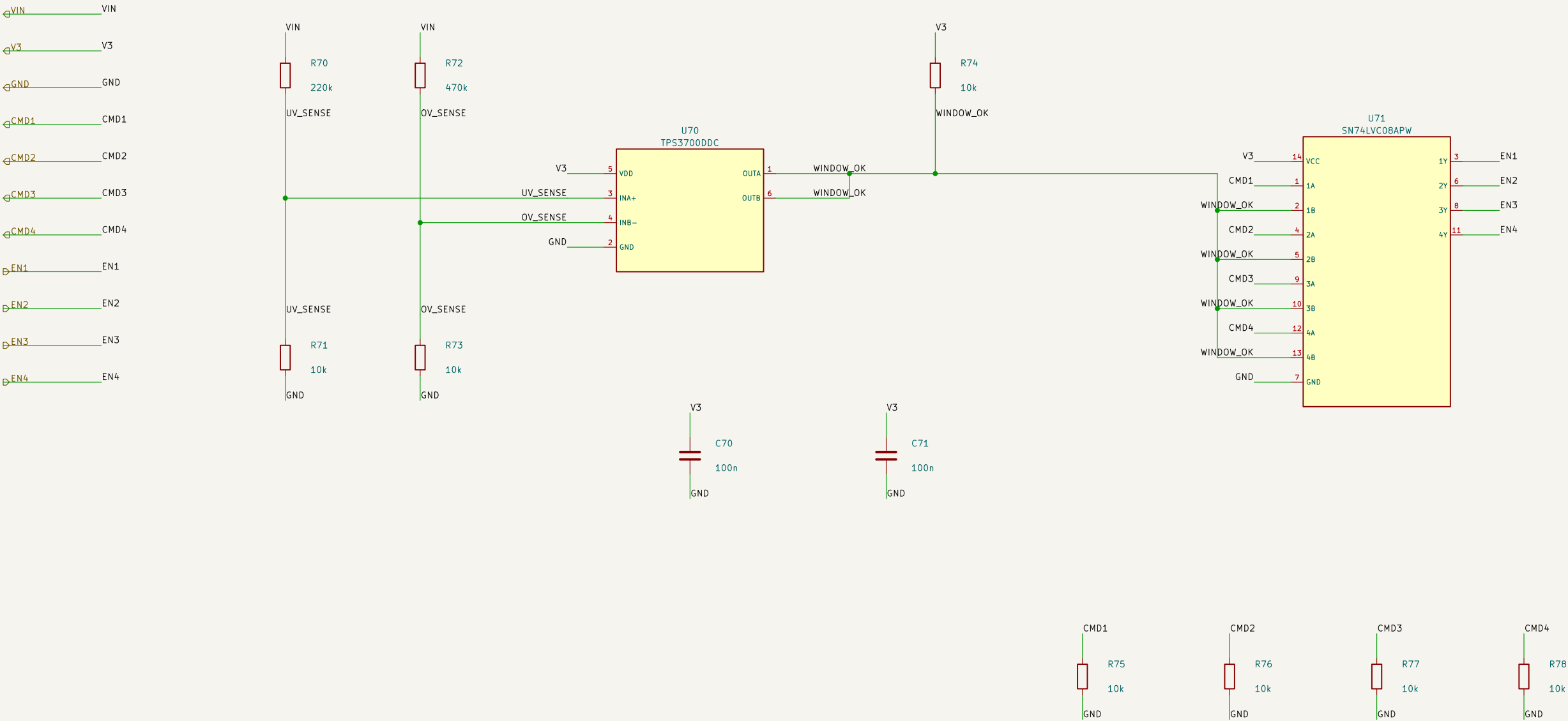


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ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
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TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



DC PROTECTION | one hardware voltage window | four independent default-off commands



TPS3700: 9.2 V UV rising / 19.2 V OV rising nominal. Open-drain outputs wired together.
74LVC08: ENi = CMDi AND WINDOW_OK; each command has a 10k pulldown.
Each outlet also has an EN pulldown: reset, missing logic supply and disconnected command request OFF.
A voltage recovery can re-enable an asserted command and reset a latched breaker; firmware must deassert on unexpected PG loss.
Shared inhibit is a common failure point, not a redundant safety channel. Local eFuse OVLO remains independent.
This circuit does not disconnect VIN from the regulator, heater stages or eFuse input pins.